Cantera vs NN | Test IC 04 | NRMSE [%]

T0 = 1317.5 K P0 = 1.244 atm phi0 = 1.984 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	2.09	4.55	10.8	18	0.448	42.5	3.54	0.413	1.47	N/A	Complete
2 P-only DeepONet10	**10k**	1.69	0.88	3.02	1.68	0.186	0.792	0.222	0.573	0.937	N/A	Complete
2 P-only DeepONet10	**long**	0.394	0.15	0.279	0.469	0.138	2.82	0.239	0.176	0.28	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	1.91	1.07	3.57	1.99	0.204	1.58	0.292	0.43	0.705	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	0.142	0.0889	0.211	0.201	0.039	1.53	0.104	0.1	0.138	N/A	Complete
4 T/P DeepONet10	**10k**	3.65	0.816	4.14	3.02	0.403	11.1	0.928	0.931	0.984	N/A	Complete
4 T/P DeepONet10	**long**	29.7	7.02	30.3	40.7	0.323	161	11.1	1.51	370	N/A	Complete
5 ANN10 + T/P DeepONet10	**10k**	3.08	0.662	3.39	2.06	0.378	10.8	0.842	0.646	0.637	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	21.4	4.5	17.5	21.7	0.14	163	10.4	0.323	2.07	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	13.8	3.79	17.2	6.25	0.921	49.8	2.22	5.91	83.6	N/A	Complete
6 T/P/ϕ DeepONet10	**long**	22.5	7.15	31.8	14.9	0.464	37.6	2.84	1.02	7.26	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.908	0.267	1.08	0.408	0.166	3.2	0.265	0.445	0.374	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	19.6	6.75	28.7	10.3	0.305	37.6	2.45	0.801	6.42	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.